

切削参数参考表
Recommended Milling Conditions

碳素钢 Carbon Steel P

合金钢 Alloy Steel P

调质钢 Prehardened Steel P

高硬度钢 Hardened Steel H

高硬度钢 Hardened Steel H

高硬度钢 Hardened Steel H

铜合金 Copper N

长颈平底 Long Neck Square

加工材料 Work Material		高速钢 High Speed Tool Steels SKH (~65HRC)				淬火钢 Hardened Steels SKD11 · PD613 (~62HRC)				淬火钢 Hardened Steels HPM-38 · STAVAX · SKD61 (~55HRC)				碳素钢 · 调质钢 Carbon Steels · Prehardened Steels S50C · NAK55 · NAK80 · HPM-1 (~43HRC)			
外 径 Dia.	颈长(l1) Under Neck Length	主轴 转速 Spindle Speed	进给 速度 Feed	主轴 转速 Spindle Speed		主轴 转速 Spindle Speed	进给 速度 Feed	切深量 Depth of Cut		主轴 转速 Spindle Speed	进给 速度 Feed	切深量 Depth of Cut		主轴 转速 Spindle Speed	进给 速度 Feed	切深量 Depth of Cut	
		min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm
0.1	0.3	40,000	70	0.002	0.03	40,000	100	0.002	0.04	40,000	120	0.003	0.05	40,000	150	0.005	0.06
	0.5	40,000	50	0.002	0.03	40,000	60	0.002	0.04	40,000	80	0.003	0.05	40,000	100	0.005	0.06
	0.75	40,000	30	0.001	0.03	40,000	50	0.001	0.04	40,000	70	0.002	0.05	40,000	80	0.003	0.06
	1	40,000	20	0.001	0.03	40,000	40	0.001	0.04	40,000	50	0.001	0.05	40,000	60	0.002	0.06
	1.5	40,000	10	0.001	0.03	40,000	30	0.001	0.04	40,000	30	0.001	0.05	40,000	40	0.001	0.06
0.15	0.5	40,000	80	0.002	0.04	40,000	100	0.002	0.06	40,000	120	0.003	0.07	40,000	150	0.005	0.09
	0.75	40,000	60	0.001	0.04	40,000	80	0.001	0.06	40,000	100	0.002	0.07	40,000	120	0.003	0.09
	1	40,000	40	0.001	0.04	40,000	60	0.001	0.06	40,000	80	0.001	0.07	40,000	100	0.002	0.09
	1.5	40,000	20	0.001	0.04	40,000	40	0.001	0.06	40,000	60	0.001	0.07	40,000	80	0.002	0.09
	2	40,000	10	0.001	0.04	40,000	20	0.001	0.06	40,000	40	0.001	0.07	40,000	60	0.001	0.09
0.2	0.5	30,000	120	0.003	0.06	30,000	160	0.003	0.08	30,000	200	0.003	0.1	30,000	240	0.005	0.12
	1	30,000	80	0.003	0.06	30,000	120	0.003	0.08	30,000	150	0.003	0.1	30,000	180	0.005	0.12
	1.5	30,000	60	0.002	0.06	30,000	80	0.002	0.08	30,000	100	0.002	0.1	30,000	120	0.003	0.12
	2	30,000	30	0.002	0.06	30,000	40	0.002	0.08	30,000	50	0.002	0.1	30,000	80	0.003	0.12
	2.5	25,000	30	0.001	0.06	25,000	40	0.001	0.08	30,000	50	0.001	0.1	30,000	60	0.002	0.12
0.3	1	22,000	20	0.001	0.06	25,000	30	0.001	0.08	25,000	40	0.001	0.1	30,000	40	0.002	0.12
	4	22,000	10	0.001	0.06	25,000	20	0.001	0.08	25,000	20	0.001	0.1	30,000	20	0.002	0.12
	1	30,000	200	0.003	0.09	30,000	250	0.003	0.12	30,000	300	0.003	0.15	30,000	350	0.007	0.18
	1.5	30,000	120	0.003	0.09	30,000	160	0.003	0.12	30,000	200	0.003	0.15	30,000	260	0.007	0.18
	2	25,000	100	0.003	0.09	30,000	120	0.003	0.12	30,000	150	0.003	0.15	30,000	180	0.005	0.18
0.4	2.5	20,000	60	0.002	0.09	25,000	80	0.002	0.12	25,000	100	0.002	0.15	30,000	150	0.004	0.18
	3	20,000	30	0.002	0.09	25,000	40	0.002	0.12	25,000	50	0.002	0.15	30,000	70	0.004	0.18
	4	20,000	20	0.001	0.09	25,000	20	0.001	0.12	25,000	30	0.001	0.15	30,000	20	0.003	0.18
	6	20,000	10	0.001	0.09	25,000	10	0.001	0.12	25,000	10	0.001	0.15	30,000	10	0.003	0.18
	1	25,000	300	0.005	0.12	30,000	350	0.005	0.16	30,000	400	0.005	0.2	30,000	450	0.01	0.24
0.5	1.5	25,000	250	0.005	0.12	30,000	330	0.005	0.16	30,000	360	0.005	0.2	30,000	400	0.01	0.24
	2	25,000	220	0.005	0.12	25,000	280	0.005	0.16	30,000	320	0.005	0.2	30,000	360	0.01	0.24
	3	18,000	180	0.003	0.12	20,000	220	0.003	0.16	25,000	260	0.004	0.2	30,000	320	0.008	0.24
	4	18,000	120	0.002	0.12	20,000	160	0.002	0.16	25,000	200	0.003	0.2	30,000	250	0.006	0.24
	6	16,000	70	0.001	0.12	18,000	130	0.002	0.16	22,000	150	0.002	0.2	25,000	200	0.004	0.24
0.6	1	20,000	400	0.005	0.15	23,000	450	0.007	0.2	25,000	500	0.01	0.25	30,000	550	0.02	0.3
	2	20,000	320	0.005	0.15	23,000	380	0.007	0.2	25,000	420	0.01	0.25	30,000	500	0.02	0.3
	3	20,000	280	0.003	0.15	23,000	320	0.005	0.2	25,000	350	0.007	0.25	30,000	420	0.015	0.3
	4	20,000	200	0.002	0.15	23,000	240	0.003	0.2	25,000	280	0.005	0.25	25,000	380	0.01	0.3
	6	16,000	100	0.002	0.15	18,000	150	0.002	0.2	20,000	200	0.003	0.25	25,000	300	0.005	0.3
0.7	8	12,000	60	0.001	0.15	14,000	130	0.002	0.2	16,000	160	0.002	0.25	20,000	200	0.005	0.3
	2	20,000	350	0.005	0.18	23,000	400	0.007	0.25	25,000	500	0.01	0.3	30,000	550	0.02	0.35
	3	20,000	300	0.003	0.18	23,000	350	0.005	0.25	25,000	450	0.007	0.3	30,000	500	0.015	0.35
	4	20,000	250	0.002	0.18	23,000	300	0.003	0.25	25,000	400	0.005	0.3	25,000	450	0.01	0.35
	6	16,000	150	0.001	0.18	18,000	200	0.002	0.25	20,000	300	0.002	0.3	25,000	350	0.001	0.35
0.8	8	16,000	100	0.001	0.18	18,000	150	0.001	0.25	20,000	200	0.002	0.3	25,000	300	0.001	0.35
	10	16,000	80	0.001	0.18	18,000	100	0.001	0.25	20,000	100	0.002	0.3	25,000	200	0.001	0.35
	4	20,000	320	0.007	0.21	23,000	400	0.015	0.28	25,000	560	0.02	0.35	25,000	690	0.03	0.4
	6	16,000	240	0.007	0.21	18,000	300	0.012	0.28	20,000	410	0.015	0.35	25,000	550	0.02	0.4
	8	12,000	180	0.005	0.21	14,000	230	0.007	0.28	16,000	330	0.01	0.35	20,000	430	0.012	0.4
0.9	10	10,000	120	0.002	0.21	12,000	180	0.003	0.28	13,000	200	0.005	0.35	16,000	300	0.008	0.4

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外 径 Dia.	颈长(t1) Under Neck Length	主轴 转速 Spindle Speed	进给 速度 Feed	主轴 转速 Spindle Speed		主轴 转速 Spindle Speed	进给 速度 Feed	切深量 Depth of Cut		主轴 转速 Spindle Speed	进给 速度 Feed	切深量 Depth of Cut		主轴 转速 Spindle Speed	进给 速度 Feed	切深量 Depth of Cut	
		min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm
0.8	2	20,000	550	0.012	0.24	23,000	650	0.03	0.32	25,000	780	0.04	0.4	25,000	850	0.05	0.45
	3	20,000	550	0.012	0.24	23,000	650	0.02	0.32	25,000	780	0.03	0.4	25,000	850	0.04	0.45
	4	20,000	500	0.007	0.24	23,000	600	0.015	0.32	25,000	700	0.025	0.4	25,000	800	0.03	0.45
	6	16,000	350	0.005	0.24	18,000	450	0.01	0.32	20,000	550	0.02	0.4	20,000	620	0.025	0.45
	8	12,000	200	0.003	0.24	14,000	300	0.005	0.32	16,000	400	0.007	0.4	16,000	500	0.015	0.45
	10	10,000	150	0.003	0.24	12,000	180	0.005	0.32	16,000	350	0.007	0.4	16,000	400	0.012	0.45
	12	10,000	120	0.002	0.24	12,000	120	0.003	0.32	13,000	220	0.005	0.4	16,000	300	0.007	0.45
1	4	14,000	500	0.02	0.3	18,000	800	0.03	0.4	23,000	700	0.04	0.5	25,000	1,000	0.05	0.6
	6	10,000	400	0.007	0.3	14,000	600	0.01	0.4	18,000	600	0.02	0.5	20,000	900	0.03	0.6
	8	8,000	340	0.005	0.3	12,000	500	0.01	0.4	16,000	500	0.02	0.5	18,000	800	0.03	0.6
	10	6,000	250	0.005	0.3	10,000	400	0.007	0.4	14,000	400	0.01	0.5	16,000	600	0.02	0.6
	12	6,000	180	0.004	0.3	10,000	300	0.005	0.4	13,000	320	0.01	0.5	16,000	500	0.02	0.6
	16	5,500	150	0.003	0.3	9,000	250	0.004	0.4	12,000	320	0.006	0.5	14,000	400	0.012	0.6
1.5	6	14,000	500	0.02	0.45	18,000	700	0.03	0.6	20,000	800	0.04	0.75	23,000	1,000	0.06	0.9
	8	10,000	380	0.01	0.45	14,000	600	0.03	0.6	18,000	600	0.03	0.75	20,000	900	0.06	0.9
	10	10,000	350	0.01	0.45	14,000	500	0.02	0.6	16,000	500	0.03	0.75	20,000	800	0.04	0.9
	12	8,000	310	0.007	0.45	12,000	430	0.02	0.6	14,000	500	0.02	0.75	16,000	700	0.04	0.9
	16	6,800	200	0.005	0.45	9,000	300	0.007	0.6	10,000	360	0.01	0.75	12,000	500	0.02	0.9
2	8	9,500	500	0.02	0.6	12,000	600	0.04	0.8	16,000	800	0.05	1	18,000	900	0.07	1.2
	10	9,500	450	0.02	0.6	12,000	500	0.04	0.8	14,000	700	0.05	1	16,000	800	0.06	1.2
	12	8,200	400	0.01	0.6	10,000	500	0.03	0.8	12,000	600	0.04	1	14,000	700	0.05	1.2
	16	7,500	340	0.007	0.6	9,200	400	0.02	0.8	10,000	500	0.03	1	12,000	600	0.04	1.2
备注 Notes		※本切削参数仅供参考。请根据实际的加工形状及使用机床等调整切削参数。 ※切深量的ap表示轴向切入量，ae表示步距量。 ※加工淬火钢时，建议使用油雾冷却方式。 ※轴向进刀建议采用螺旋进刀及倾斜进刀方式。 ※ t1(颈长) / D(刀径)超过8倍时，立面附近的进给速度须调整至50%以下，切深量：ae调整至30%以下。 ※沟槽切削时建议参考切削参数表，切深量ap及进给速度设定为50%以下，采用来回切削加工方式。 ※发生振刀时，请以相同的比率降低主轴转速和进给速度。此外，主轴转速过低时，也以相同的比率调。															
		※These recommended cutting conditions indicate just reference. It should be adjusted according to milling shape and machine type ※ap:Axial Depth of Cut, ae:Radial Depth of Cut. ※Recommend to use oil mist coolant for machining hardened steel. ※Recommend to apply herical or ramping for approaching into axial direction. ※ Adjust feed rate 50% lower and cutting depth(ae) 30% lower for milling deep wall area. Under Neck Length t1/D exceeds 8 for stable milling ※For slotting, recommend reciprocating milling by adjusting feed & ap in below 50% of recommended milling condition. ※Reduce both spindle speed and feed at same rate for chattering and also for insu cient spindle speed of a machine.															